

Score-linked practice intelligence from long-form violin training video

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High-level music training produces dense evidence: daily practice videos, repeated repertoire, changing technical constraints, score-free technique exercises, and periodic performance goals. Most of that evidence is difficult to use because it is stored as unstructured media rather than as a score-linked or pattern-linked longitudinal record. We report Curtis, a live AO Labs system that converts messy long-form violin practice videos into structured, evidence-linked practice records using active practice detection, playable audio/video evidence windows, score alignment targets, technique-pattern targets, heat-map targets, repertoire tracking, and passage-level performance observation. The checked live source state at 2026-05-15 05:30 PM EDT used the YouTube channel @nalalan as the primary source, indexed 219 public videos, marked 112 as practice candidates, and stored 106 active-scan sample results. The uploaded-video ledger begins at violin 1, uploaded on 20 December 2025, and currently counts 48 violin-numbered or date-titled practice logs totaling 234 h 57 min of public session duration; this duration is not counted as active practice time. The daily-record state contains 41 indexed practice days, 3 audio-evidence records, 2 score-audio-only records, 1 detected-note-fragment record, 94 active-violin scan sample results, and 12 checked non-violin results. Total practice time is counted only from media windows where violin playing is detected; it is not tied to transcription or score matching. The live active-practice scan has persisted 331 active violin-playing sub-intervals across the stored media samples, totaling 54 min 2 s of scan-derived active audio; the coverage ledger now reports 2 h 21 min measured active practice from 2 h 28 min of checked video after the scan supersedes noisy transcription-duration evidence where scan results exist. The first chronological violin 1 window, *0-90, has been checked as non-playing/unclear; the windows *90-1260, *1350-1530, and *1620-2250 were uploaded from owner media and converted into active-practice evidence; *1260-1350 and *1530-1620 have been checked as non-playing/unclear; the next chronological pending window is *2250-2340. The current 223 h 5

min archive-wide estimate remains provisional because the checked windows are still sparse and biased toward captured samples; it is not the final full-archive practice total. All unverified room, setup, laptop, silence, or non-playing samples are withheld rather than counted as measured practice evidence. This PDF is now also the progress tracker for the long-phrase transcription and exact score-matching goal: every meaningful Curtis transcription, score, practice-time, heat-map, repertoire, or observation improvement should update this paper in the same work cycle. The primary interface has three top-level surfaces: a paper card, analyzed practice days, and repertoire, with a bottom transcription-completion tracker that reports a 100-point implementation gate score, precise weighted progress, completed work, remaining gates, and the next scan action. The analyzed-days evidence row reports total practice time first, then video checked for practice-time detection, then the full uploaded-video archive so the interface does not hide either the multi-hour corpus or the scan coverage. The analyzed-days list keeps practice days chronological and displays exact audio-checked D4 and A4 fragments from the 2026-05-03 Scherzo-Tarantelle practice sample as detected-note evidence, not as score-location matches. Note-match groups are now driven by pitch-class sequence agreement against an available score-derived sequence or a separately labeled reference-audio sequence; the symbolic-score path parses MusicXML into solo-violin notes and accepts a displayed score phrase only when the detected MIDI sequence, score MIDI sequence, written spelling context, visible score noteheads, visible range, and an actual source-score crop all agree. Source-specific acceleration targets can create pending measure-level or phrase-level candidates when the score-side notes come from source-backed MusicXML and local playable media exists, but those candidates do not become accepted score evidence until the visual score panel is a non-generated source-score crop for the exact accepted range. Generated notation may render the transcription side, but it cannot appear as the accepted score snippet. The Scherzo-Tarantelle target now has a local source PDF, a seven-note source-backed symbolic opening phrase D C Bb D C Bb D, rejected source-crop review fixtures, a 951-glyph local score-map queue, a 214-item unaccepted notehead hypothesis queue, 8 staff-level source review packets, and a separate 179-note local reference-audio pitch trace; loose reference-audio matches alone stay hidden from accepted score evidence. Each daily record withholds embedded playback, paired audio, and practice-time credit until violin playing is detected in the source window, withholds broader sheet-music notation until reliable note extraction supports either score alignment or a score-free technique-pattern record, and withholds exact score-location snippets until a played window has explicit score/transcription

visual agreement and actual-source score agreement rather than only a source-confirmed piece title or broad pitch-sequence match. Single-pitch score crops are withheld unless the highlighted score note has explicit visual note verification. All current Scherzo-Tarantelle single-pitch crops and the prior five-note 5/3 crop are rejected after review because the visible score noteheads did not match the displayed transcription. The live system therefore has no accepted score crop; detected D4/A4 audio fragments remain audio-checked transcription evidence only, and the five-note Scherzo phrase remains a pending symbolic candidate rather than accepted score evidence. Draft notation does not render as transcription; repeated single-pitch traces such as a spurious D D D D stream remain out of final score-verified notation. The transcription-run PDF now lists all stored detected note series for the day, while displayed score phrases require MIDI-level note agreement, exact score/transcription visual agreement, visible notehead agreement, and an actual source-score crop; looser reference-audio candidates remain hidden from accepted score evidence. Curtis keeps source-confirmed pieces, explicit calibration titles such as scales and arpeggios, official YouTube Data API public-reference metadata, nearest reference targets, and score context available without turning untrusted note events into accepted score evidence. Manual day-level source labels can be entered from the analyzed-day record, but these labels only name the practice source; they do not unlock score snippets, heat maps, readiness scores, or repertoire progress without matching audio and score evidence. Public labeled YouTube references seed likely reference targets but do not become repertoire, audio fingerprints, or judged performance evidence until a permitted media path and score or pattern alignment exists. Active material with no confirmed score is labeled as piece-or-exercise pending rather than forced into a wrong repertoire guess. Repertoire entries are promoted only from confirmed daily-record evidence and keep possible labels, user-rejected false labels, weak evidence, and readiness scores inferred only from piece identification separate from confirmed repertoire and judged playing quality. The system uses GPT-5.5 for vision and text review and separate audio models for identification and verification, while preserving explicit limits: Curtis admission cannot be predicted from current samples, weak audio/video evidence is marked unjudged, whole-session transcription is not yet complete, and the system reports practice signals rather than admissions probability. The contribution is an experimental method for making practice visible, searchable, measurable, and tied to actual musical evidence.

1 Introduction

Elite music preparation depends on repeated, deliberate practice and feedback over long time horizons (1, 2). The useful data are not only polished performances. They include ordinary practice sessions, failed takes, camera angle limitations, recurring technical problems, and changes in focus across days. A violinist recording daily practice can therefore produce a large longitudinal dataset, but the raw dataset does not automatically become an operating record.

Curtis addresses this problem as a live software system rather than as a portfolio page (3). The system indexes public practice media, separates uploaded video duration from detected violin-playing practice time, records technical observations, and keeps evidence tied to clips and generated notation. It is designed for the Curtis Institute goal, but it does not claim to estimate an acceptance probability. The current role of the system is narrower: observe practice evidence, retain history, separate judged from unjudged evidence, and identify the specific passage-level blockers that can be tested in subsequent recordings.

The core pipeline is video → active practice detection → clips → music transcription → repeat grouping → score matching → heat map → repertoire update → passage-level observations. The output is intentionally small: analyzed practice days and an automatically updated repertoire list, both grounded in clips, notation, active playing time, and confidence limits.

The system is also part of AO Labs' broader progress architecture. In May 2026, AO Progress was introduced as a shared ledger for cross-application state and long-term changes across public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state (4). Curtis supplies one stream in that ledger: timestamped media and review evidence for musical development.

2 System state

The production state uses the live backend at <https://curtis.aolabs.io/api/curtis/media-status>. The checked source state at 2026-05-15 05:30 PM EDT reported the values in Table 1. These are operating facts, not outcome claims.

A later implementation check at 2026-05-14 08:29 AM EDT corrected and extended the Scherzo-Tarantelle score-truth source. The previous D A# G D source-score phrase is rejected as an accepted symbolic-score match because the local IMSLP solo-violin score shows the opening source phrase as

D C Bb D C Bb D. The 2026-05-03 Scherzo-Tarantelle phrase reports detected notes A#4 D5 C5 119
A#4 D5; for display it is score-spelled as Bb4 D5 C5 Bb4 D5 against the same source score notes, 120
but it is now a pending symbolic candidate rather than accepted score evidence. The 2026-05-14 121
07:04 PM EDT actual-source score-gate update added an exact-range source crop for that five-note 122
subrange and prohibited generated notation SVGs from appearing as accepted score snippets. The 123
2026-05-14 09:45 PM EDT spelling-context update closed a separate visual mismatch by carrying 124
the source key signature into the transcription panel and requiring visual-range and score-spelling 125
agreement. The 2026-05-14 10:52 PM EDT visual-crop demotion then rejected the same source crop 126
because Alan's review showed that the visible source-score notes still did not match the transcription. 127
Generated notation remains available only as the transcription-side rendering; a score panel must use a 128
non-generated source-score image whose visible noteheads, register, spelling context, parsed score-note 129
range, and explicit visible-note-sequence verification all match the transcription. The 2026-05-15 05:00 130
PM EDT strict evidence-gate update makes that demotion enforceable in code: a score crop cannot 131
count as visible accepted evidence unless score box centers, visible score notes, written spelling, exact 132
note-plus-octave sequence, audio-to-transcription agreement, transcription-to-score agreement, and an 133
accepted truth-evidence flag all pass. The 2026-05-14 02:51 PM EDT target-triage update checks 134
four-note-or-longer reference-audio targets against the current verified symbolic score excerpt before 135
any promotion attempt. In the current live state, the top source target is D D D# D D A# G; the verified 136
score map is D C A# D C A# D; the best contiguous overlap is D, or 1/7 notes, so the target is classified 137
as `source_score_sequence_not_found`, not accepted score evidence. The 2026-05-14 03:49 PM 138
EDT score-map workbench update added a local source-map verification queue with 779 unaccepted 139
score glyph candidates across 7 staves from page 2 of the local IMSLP solo part. The 2026-05-14 04:24 140
PM EDT note-hypothesis update filters that queue into 172 unaccepted likely notehead hypotheses with 141
staff-position pitch labels. The 2026-05-14 05:17 PM EDT review-packet update fixed the skipped-staff 142
failure mode in the source detector, expands page-2 coverage to 951 unaccepted score glyphs and 143
214 unaccepted note hypotheses across 8 staves, and writes 8 staff-level source review packets. The 144
2026-05-15 12:45 AM EDT truth-workbench update adds a persistent accepted/rejected/pending 145
truth-item path and a compact site surface for queued source targets, accepted truth clips, rejected 146
mismatches, and regression benchmarks; it also makes the visible-note-sequence flag mandatory for 147
score-visual and actual-source score locks. The deployed strict-gate tracker reports 43.126% exact 148
weighted progress, with 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned 149
windows, 0 score-coordinate heat-map fragments, 0 score-visual locks, 0 actual-source score locks, 0 150

accepted truth items, 2 source-verification targets queued for truth review, 14 pitch-sequence groups, 2 source-target score checks, 0 verified source targets, 7 verified symbolic score notes, 0 verified source snippets, 951 unaccepted score glyph candidates, 214 unaccepted score-note hypotheses, and 8 staff review packets. This is implementation progress only; a five-note candidate, rejected source-target checks, queued truth checks, and unaccepted score-glyph, note-hypothesis, or review-packet candidates do not solve full long-phrase or full-session transcription.

2.1 Media indexing

The source inventory stores platform, title, publication time, duration, view count, candidate reasons, media kind, and blocker state. Dated long-form practice logs are treated as high-value longitudinal records because the title contains a practice date and the duration preserves session-level evidence. The uploaded-video ledger starts at the violin 1 marker and counts only violin-numbered or date-titled practice logs, so unrelated long public videos are retained in inventory without being silently counted as violin practice. This ledger is not the active-practice total. Active practice is measured only after media processing identifies windows where violin playing is actually occurring.

2.2 Review evidence

The review layer stores section-level findings with a dimension, evidence source, judgment, and practice constraint. The current dimensions include tone, intonation, shifts, time, articulation, and audition delivery. Findings can be strong, needing work, or unjudged. The unjudged state is part of the system design: if the sound is not available, the hand is obscured, the camera angle hides a transition, or still frames do not support a timing claim, the system should not convert weak evidence into a confident critique.

3 First readout

The first live readout is experimental and intentionally conservative. The strongest immediate value is that the system has already converted a large, timestamped practice archive into a queryable corpus with captured media samples, stored section reviews, and title-confirmed practice records. The home screen has three top-level surfaces: a paper card, analyzed practice days, and repertoire, followed by a compact transcription-completion tracker at the bottom. The first evidence row now reports total

Table 1. Curtis source state checked at 2026-05-15 05:30 PM EDT.

Signal	Current value
Primary source	YouTube channel @nalalan
Inventory	219 indexed videos
Practice candidates	112 videos
Long-form candidates	92 videos
Uploaded-video marker	violin 1, uploaded 20 December 2025
Uploaded-video ledger	48 videos, 234 h 57 min uploaded session duration
Estimated total practice time	223 h 5 min extrapolated from the checked-window active-playing ratio; provisional because the checked windows remain sparse and sample-biased
Measured practice time	2 h 21 min of detected violin-playing footage; this is independent from transcription and cannot exceed checked video coverage
Video checked for practice time	2 h 28 min of the 234 h 57 min uploaded-video ledger
Active-practice scan	331 active sub-intervals persisted from stored media samples; 54 min 2 s of scan-derived active audio; 106 sample results; violin 1 windows through *2160–2250 checked; 250 queued windows still pending owner media
Unchecked archive duration	232 h 28 min still awaiting violin-playing detection
Media samples	106 active-scan sample results; 94 active violin; 12 checked non-violin
Reviewed sections	80 sections
Curtis-focused findings	80 findings
Latest dated practice log	5-6-26
Review model	GPT-5.5 for vision/text, separate audio models for audio evidence and piece verification
Confirmed labels	5/1 Haydn Symphony No. 94 IV; 5/2 and 5/3 Wieniawski Scherzo-Tarantelle
Training sources	3 source-confirmed practice sources
Score-reference targets	3 configured
Reference training	3 source-confirmed score targets, 22 pitch/rhythm windows, title-labeled calibration sources for future scale/arpeggio uploads, and 4 official YouTube Data API public-reference queries kept separate from repertoire claims and audio evidence
Analyzed practice days	41 indexed days; 3 practice-time measured daily records; 3 audio-evidence records; 2 score-audio-only records; 1 detected-note-fragment record; 3 confirmed source labels, score targets, and a source-backed Wieniawski seven-note symbolic opening phrase
Machine pitch events	Detected-active-section onset-aware pYIN plus onset-bounded spectral pitch rescue for fast passages, audio-agreement counts, pitch-sanity filtering, repeated-pitch-collapse guarding, practice-coverage fields, title-labeled calibration handling, and fingerprints stored only when quality gates pass; broad candidate micro-transcription is withheld, while detected note series are preserved in the transcription PDF and promoted to note-match groups only after pitch-class sequence agreement with an available score-derived or reference-audio sequence
Specific observations	Rendered only when grounded in reliable score-linked evidence or repeated score-free technique-pattern evidence
Note-match windows	No accepted Scherzo-Tarantelle score match is currently displayed after the 2026-05-03 five-note crop was demoted; detected notes A#4 D5 C5 A#4 D5 remain a symbolic phrase candidate only because the visible notes in the actual source-score crop did not match the transcription; longer phrase and rhythm transcription remain unsolved
Transcription solve tracker	43.126% live exact weighted progress after strict evidence-gate update; 0 accepted long phrases; 0 accepted measure matches; 0 exact score-aligned windows; 0 score-coordinate heat-map fragments; 0 score-visual locks; 0 actual-source score locks; 0 accepted truth items; 2 source-verification targets queued for truth review; 2 source-target score checks; 0 verified source targets; 7 verified symbolic score notes; 0 verified source snippets; 951 unaccepted score glyph candidates; 214 unaccepted score-note hypotheses; 8 staff review packets
Daily evidence surface	Local sample playback only for violin-positive windows, notation withheld until reliable note/rhythm verification, score snippets hidden until played-to-score agreement, and explicit whole-session transcription limit
Curtis-level completion	Not scored when the available evidence only identifies repertoire
Current focus	Capture a playable audition-view take
Current constraint	Record 30 seconds of sustained tone
Boundary	Curtis admission cannot be predicted from current samples

practice time first, then the video already checked for violin-playing, then the 234 h 57 min YouTube practice archive; this prevents the public practice ledger from being confused with either transcription coverage or uploaded-video duration. Each analyzed day groups same-day videos, preserves uploaded duration, and names confirmed or uncertain piece evidence without promoting guesses. Playable local media samples, paired audio/video evidence windows, and practice-time credit are now withheld unless violin playing is detected in the source window. The owner-sync sampler now reuses locally cached captured media, uploads cached violin-positive windows before blind new captures, resumes interrupted queue-only runs from cached pending-window media including non-playing windows, consumes active-scan pending windows before generic expansion or blind sampling, has a queue-only mode for chronological pending-window coverage, expands around already-found violin-positive windows, and can switch into full-check mode to walk every 90-second window in order instead of staying limited to sampled windows; the live analyzed-days surface currently exposes violin-playing practice-time evidence for 5/1, 5/2, and 5/3, and the first twenty-five violin 1 windows are now represented as checked coverage, with three non-playing/unclear windows and twenty-two active windows. Broad sheet-music notation is also withheld until reliable note/rhythm verification, so audio evidence cannot appear as a failed transcription. The current visible notation is limited to exact audio-checked D4 and A4 fragments; the previously displayed actual-source-score-locked five-note Scherzo-Tarantelle phrase from 5/3 is demoted because the visible source crop did not show the same notes as the transcription. Single-note Scherzo-Tarantelle score crops remain rejected. A multi-note score panel now renders only when the detected MIDI sequence, the parsed score MIDI sequence, the written spelling context, and a non-generated source-score crop agree on the same visible range; generated notation SVGs are no longer allowed to stand in for the score panel. A source-confirmed piece title creates a score target, and a detected pitch-class sequence match can show the played clip and transcription, but broad score-region estimates and visually unverified crops do not render as sheet-music snippets. The daily record includes a compact source-label field so a missing or wrong day title can be corrected without letting that correction masquerade as score alignment. The analyzed-days surface no longer promotes broad candidate transcription while candidate micro-transcription fails audible match review; those windows remain hidden support data or audio evidence only. Active scans with no confirmed piece are treated as possible repertoire or score-free technique exercise evidence rather than as failed score matching. The repertoire surface is evidence-backed: an entry appears only when a daily record carries confirmed source evidence, and each entry keeps practice dates, clips, score targets, practice time when measured, and current blocker limits. Generic etude/caprice evidence

remains outside the piece-title field, and possible or uncorroborated repertoire labels remain checkable against the source window. These statements are not generic pedagogy; they are extracted from stored review findings, sample indexes, source corrections, public-domain score targets, pitch-quality checks, and the uploaded-video ledger.

Table 2. Current evidence classes and limits.

Class	Stored signal	Limit
Audio	Tone, intonation, shifts, time, articulation	Some audio reviews fail or cannot inspect the excerpt cleanly
Video	Setup, posture, bow lane, visible hand/bow evidence	Still frames can hide motion, sound, and transition accuracy
Metadata	Dates, duration, channel, view count, practice-candidate labels	Metadata cannot judge playing quality by itself
Progress plan	One focus, practice constraint, short session plan	The plan is a next experiment, not an outcome prediction

Current transcription limit. Machine note output now has three display gates. First, the media window must be classified as violin-positive before it can appear as playable evidence or count toward practice time. Second, the note stream must audibly match the paired audio before any notation can render. Third, repertoire passages need score alignment, while score-free exercises need repeated-pattern grouping before they can become full practice transcription. Source-confirmed score targets are also display-gated: broad score snippets are hidden unless the corresponding played window has explicit score-location agreement, and single-pitch score crops require explicit visual note verification before they can render. Score-note sequences are now source-gated before matching: a scanned PDF plus guessed scorePitchClassSequences is ignored unless the note sequence has an explicit symbolic-score source, manual source confirmation, MusicXML/source-confirmed status, or exact score-location metadata. Curtis now includes a symbolic-score parser for inline MusicXML, local .musicxml, and local .mxl targets. That path extracts score notes, measures, pitch classes, written note names, octave-aware MIDI values, and duration fields. The parser now preserves written flats such as B-flat for score staff placement while matching their enharmonic pitch class to detected audio, so a source-score B-flat can no longer be rendered as an A-sharp staff position. The Scherzo-Tarantelle target now carries a source-backed seven-note symbolic opening phrase from the local IMSLP solo part, but the current source-score crops are rejected review material rather than

accepted score evidence because visual inspection showed that the displayed crop did not match the transcribed note sequence. A visible score match normally requires at least five contiguous played pitch classes and at least three distinct pitch classes; target-specific source excerpts may still use a four-note measure-level acceleration gate, but no current Scherzo-Tarantelle phrase is counted as accepted long-phrase progress after the crop demotion. One-note overlaps and repeated single-pitch runs are still rejected as score evidence. When the symbolic gate passes, generated notation can render the transcription, but the score panel is accepted only if it uses an actual source-score crop whose visible noteheads, register, written spelling, and exact range match the transcription; a generated SVG excerpt can no longer masquerade as the actual score. All current Scherzo-Tarantelle single-pitch crops and the prior five-note 5/3 crop are rejected and withheld because visual note review found that the highlighted or visible score notes did not match the displayed transcription; these failures are now regression fixtures. Unverified room, setup, laptop, silence, or non-playing samples remain withheld, and unverified pitch traces remain hidden support data rather than visible notation. A spectral-onset rescue path estimates pitch inside short onset-bounded violin segments when the pYIN path collapses, while the repeated-note trace guard prevents fast or complex playing from becoming a dominant repeated pitch, records the quality mode, and keeps unusable traces out of accepted score evidence. The current micro-transcription path stores candidate short contiguous runs, but it does not render them as sheet music or count them as solved transcription until the displayed notes pass paired-audio listening review. The transcription-run PDF preserves all stored detected note series for the day. Note matching uses pitch-class sequence agreement, ignores rhythm and octave for the initial grouping step, and can promote detected Scherzo-Tarantelle fragments only when the played pitch-class sequence appears in parsed symbolic score notes or another available score-derived/reference sequence; reference-audio matches alone are not treated as score notation. A separate audio-checked fragment gate can render a note when pYIN and YIN agree on the same semitone over a short stable window; the present live examples are D4 and A4, not a full passage and not a score-location match. Full-session transcription remains incomplete until every playing section in the long-form practice videos is extracted, verified as violin-positive, verified against the audio, and aligned either to a score section or to a repeated technique pattern.

4 Discussion

Curtis Media Review is intentionally conservative. It does not rank the player against applicants, infer admissions probability, or convert every video into a claim. Instead, it keeps a running technical record. The sampler now treats violin presence as a hard evidence boundary: playable clips and practice-time credit appear only after violin playing is detected in a media window, while notation waits for reliable note/rhythm verification and either score alignment or score-free pattern grouping. This makes the system useful even before it becomes musically sophisticated: it can preserve practice volume, surface recurring constraints, identify missing evidence, and make subsequent recordings easier to compare against earlier ones.

The main open problem is repertoire identification, score-free exercise representation, and evidence quality. YouTube metadata can identify candidate videos and public labeled reference sources, and the current owner-sync path can provide direct audio/video samples, but unclear camera framing, missing score/title context, improvised technical patterns, and short excerpts still prevent reliable piece naming. The backend therefore keeps generic labels out of the piece-title field, stores them as candidate evidence, requires source-window evidence for confirmed titles, and marks weak evidence explicitly. The current training ledger separates source-confirmed labels from title-labeled calibration uploads, public YouTube reference metadata, independent audio-title matches, score-free or piece-pending active windows, failed unverified machine pitch/rhythm windows, failed pitch-collapse windows, withheld candidate micro-transcription windows, audio-checked single-note fragments, detected note series, pitch-class reference matches, and true score-aligned windows. It also stores reference targets for the three confirmed source labels across two works so later passes can ask for section-level matching instead of repeating broad title guesses. The sampler classifies candidate windows using active audio, violin-range voiced pitch, onset density, pitch-class diversity, spectral features, and minimum practice-duration gates before a sample can become visible evidence. The transcription path converts sampled media to mono audio, finds audio-active playing sections, runs pitch tracking only on those sections, normalizes and harmonic-isolates each section for pitch tracking, estimates monophonic violin pitch with an onset-aware pYIN pitch track bounded to the violin range, detects onset boundaries so repeated attacks on the same pitch are not silently merged into one sustained note, runs a second spectral pitch estimate inside short onset-bounded segments when fast playing defeats pYIN, compares pitch diversity across candidate note streams, checks whether selected note events agree with an independent spectral or pYIN/onset detector, rejects out-of-range pitch artifacts, requires sustained pitch changes for pitch-track segmentation, removes very short low-confidence pitch blips,

marks isolated octave-flip corrections as uncertain, rejects repeated-pitch collapse, estimates tempo, stores detected note series for audit/PDF review, and builds pitch-class, interval, rhythm, and contour fingerprints only when quality gates pass. Daily records now carry transcription quality, material status, and practice-coverage states, and the primary interface withholds media playback unless the source window is violin-positive, withholds broad machine notation until note/rhythm verification, displays only audio-checked fragments when the stricter gate passes, hides loose reference-audio matches from the daily row, shows a source-score crop only when the highlighted score note has explicit visual note verification, and withholds broader score snippets until exact score-location alignment exists. A stronger system should align repertoire events against a symbolic or optical score representation, group score-free technical patterns by repeated note/rhythm fingerprints, identify movement sections or measure ranges when a score exists, compare against reference recordings available through permitted media, and then judge articulation, timing, tone, and tempo against the aligned passage or repeated pattern.

The broader experiment is whether a practice system can make musical development legible over time. A useful Curtis system should show whether tone, intonation, rhythm, articulation, shifts, memory, endurance, and audition delivery are improving across dated recordings. It should also preserve the source evidence behind each claim so that the dashboard does not become motivational text detached from the actual playing.

5 Roadmap to solved long-phrase transcription

This section is the working plan for making Curtis able to produce long, trustworthy audio–video–transcription–score evidence from Alan’s actual violin practice. It is intentionally part of the paper rather than a separate planning note. Whenever Curtis makes real progress toward full-session active-practice measurement, accurate transcription, score matching, heat maps, repertoire updates, or Curtis-level observation, this PDF should be updated with the new state, the acceptance evidence, and the remaining blocker.

5.1 Current tracker state

The tracker was refreshed through the 2026-05-15 05:00 PM EDT strict evidence-gate pass. The system has a real media ledger, a deployed active-practice scan route, a working owner-sync path for

captured media, a visible 100-point transcription-completion tracker with precise weighted progress, stored user-correction benchmarks, a persistent truth workbench for accepted/rejected/pending audio-score-transcription items, a local source-backed Wieniawski score PDF, a seven-note source-backed Wieniawski symbolic score opening phrase, a source-verification target queue, source-target score triage, an unaccepted score-glyph verification queue, an unaccepted score-note hypothesis queue, staff-level source review packets, and a stricter evidence gate. The 2026-05-03 Scherzo-Tarantelle five-note source crop is no longer accepted: Alan's visual review showed that the visible noteheads and note boxes did not agree closely enough with the transcription and audio evidence. Curtis now withholds source score snippets unless score box centers, visible score notes, written spelling, exact note-plus-octave sequence, paired-audio transcription agreement, transcription-score agreement, and an accepted truth item all pass together. The old A4/A5 single-note anchors remain rejected, and a one-note pitch anchor still cannot render unless the score note and octave are explicitly verified. The current top longer source target D D D# D D A# G remains unaccepted because the verified symbolic score map contains only D C A# D C A# D and gives only a one-note contiguous overlap. This is not solved long-phrase transcription yet; it removes a false five-note score lane, makes the daily-record API share the canonical active-practice coverage ledger, and keeps longer passage transcription, rhythm, repeated attempts, full-score mapping, and full archive practice-time coverage open.

Table 3. Progress tracker baseline for the long-phrase transcription goal.

Tracker item	Baseline state
Goal	Produce compact daily evidence groups: original score snippet, generated transcription, local audio, local video, heat-map context, and specific observation.
Current accepted transcription	0 score-linked transcription phrases are currently accepted. The five-note Scherzo-Tarantelle run A#4 D5 C5 A#4 D5 remains a pending audio-derived candidate, not accepted evidence.
Current accepted score match	0 actual-source score crops are currently accepted. The previous Bb4 D5 C5 Bb4 D5 crop at mm. 2-4 is rejected until score box centers, visible source notes, transcription notes, and paired audio agree under the truth gate. The current D D D# D D A# G longer source target is still rejected for now with only 1/7 contiguous source overlap; 951 local score glyph candidates, 214 staff-position notehead hypotheses, 8 staff review packets, and 2 source-verification targets are queued for truth review but are not accepted score evidence.
Current archive coverage	234 h 57 min uploaded practice video across 48 strict-ledger videos and 41 practice days; 2 h 28 min checked; 2 h 21 min measured active violin; full chronological coverage still pending.
Current estimate	223 h 5 min total active practice estimated from checked windows; provisional and sample-biased until the full chronological scan covers the archive.
Current user burden to remove	Alan is still forced to inspect whether a note is real, whether a score box points to the correct pitch, whether audio matches notation, and whether practice time means playing rather than uploaded duration.
First tracker update	This PDF now defines the full implementation plan and becomes the required progress record for future Curtis work.

Progress log

2026-05-13 11:57 AM EDT code update Backend coverage ledger, evidence-correction route, and benchmark-fixture scaffold added. Checked video time now includes sampled non-playing windows; total practice time remains measured active playing only. Live deployed coverage now reports 213 h 23 min uploaded video, 2 h checked video, 53 min 14 s measured active practice, and 211 h 23 min unchecked. Wrong score-note corrections can be stored as rejected benchmark cases. Full archive

scan and long-phrase transcription remain unsolved.

2026-05-13 12:33 PM EDT code update Low-cost active-practice scan route implemented. The new scan persists active violin-playing sub-intervals from stored violin-positive media samples, keeps checked non-playing samples separate from practice time, exposes `/api/curtis/active-practice-scan`, adds an active-scan run endpoint, queues remaining strict-ledger windows for owner media capture, and feeds persisted active intervals into daily records and the coverage ledger. Local unit tests verify active sub-interval extraction, negative-sample withholding, and coverage accounting.

2026-05-13 12:59 PM EDT deployed scan update The active-scan endpoint was deployed and run against the live Curtis state. It persisted 214 active sub-intervals from 80 stored media-sample results, representing 41 min 42 s of scan-derived active audio, and queued 250 strict-ledger windows for later owner-media capture. The coverage ledger reports 1 h 57 min 24 s measured active practice from 2 h checked and a 208 h 46 min provisional archive estimate. A regression fix now caps active practice by checked source coverage and makes active-scan results supersede noisy transcription-duration evidence where scan results exist. The full 213 h 23 min archive scan and long-phrase transcription remain unsolved.

2026-05-13 03:26 PM EDT queue update The owner-media sync now consumes the active-scan pending-window queue before choosing expansion or blind sample windows. The active-scan status endpoint can expose a bounded pending-window preview, and the run endpoint returns a compact scan/coverage payload rather than the full operations document. Regression tests now verify that queued active-practice windows are selected first and that pending-window previews preserve total pending counts while limiting returned rows. This does not solve long-phrase transcription; it closes the next M1 plumbing gap so the chronological archive scan can keep advancing without Alan manually choosing windows.

2026-05-13 03:37 PM EDT live owner-sync update A one-window owner-sync pass uploaded one cached violin-positive 5/3 sample, Njh8_zq9_DM-10725 from *10725-10815. The follow-up active scan selected that new sample and recorded one new active interval. Live coverage is now 81 active-scan sample results, 215 active sub-intervals, 42 min 9 s scan-derived active audio, 2 h 1

min 30 s checked video, 1 h 58 min 54 s measured active practice, and a 208 h 49 min provisional estimate. The first chronological pending queue item remains `violin 1` at `*0-90`; this run advanced cached-media coverage, not the chronological queue.

2026-05-13 04:05 PM EDT completion tracker and queue-only update Curtis now exposes a bottom-page transcription completion tracker backed by a 100-point implementation gate checklist. The tracker reports archive coverage, accepted long phrases, exact score windows, scan queue state, completed gates, remaining gates, and the next concrete action. A queue-only owner-sync mode was added so the active-scan pending-window list can be consumed without falling back to cached positives, expansion windows, or blind sampling. A queue-only run uploaded `otHfHMgDo2g-0` from `violin 1` at `*0-90`; the sample was classified as `not_violin_or_unclear` and therefore did not add practice time. The follow-up active scan reports 82 sample results, 72 active-violin samples, 10 checked non-violin samples, 215 active sub-intervals, 2 h 1 min 30 s checked video, 1 h 58 min 54 s measured active practice, and a 208 h 49 min provisional estimate. The next chronological pending window is `violin 1` at `*90-180`.

2026-05-13 05:44 PM EDT queue resume and practice-time ceiling update The next queue-only owner-sync attempt captured `violin 1` windows `*90-540`. The wrapper timed out after the uploads, revealing that cached pending-window media was excluded from queue-only resume. The owner-sync helper now resumes cached queue windows, including cached non-playing windows, and tests cover that path. The follow-up active scan reports 87 sample results, 77 active-violin samples, 10 checked non-violin samples, 238 active sub-intervals, 2 h 3 min checked video, 1 h 55 min 30 s measured active practice, and a 200 h 23 min provisional estimate. Practice time is now hard-capped by checked source coverage, so unchecked transcription remnants cannot make active practice exceed checked video. The next chronological pending window is `violin 1` at `*540-630`.

2026-05-13 06:01 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded `otHfHMgDo2g-540` from `violin 1` at `*540-630`. The sample was classified as `violin_positive` with a sampler score of 68.6. The follow-up active scan selected that sample, recorded 4 new active sub-intervals, and reports 88 sample results, 78 active-violin samples, 10 checked non-violin samples, 242 active sub-intervals, 2 h 4 min 30 s checked video, 1 h 57 min measured active practice, and a 200 h 32 min provisional estimate. The live accounting check passes because

active practice remains below checked video. The next chronological pending window is violin 1 at *630-720.

2026-05-13 06:26 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded otHfHMgDo2g-630 from violin 1 at *630-720. The sample was classified as violin_positive with a sampler score of 60.3. The follow-up active scan selected that sample, recorded 5 new active sub-intervals, and reports 89 sample results, 79 active-violin samples, 10 checked non-violin samples, 247 active sub-intervals, 2 h 6 min checked video, 1 h 58 min 30 s measured active practice, and a 200 h 41 min provisional estimate. The live accounting check passes because active practice remains below checked video. The next chronological pending window is violin 1 at *720-810.

2026-05-13 07:55 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded otHfHMgDo2g-720 from violin 1 at *720-810. The sample was classified as violin_positive with a sampler score of 55.2. The follow-up active scan selected that sample, recorded 3 new active sub-intervals, and reports 90 sample results, 80 active-violin samples, 10 checked non-violin samples, 250 active sub-intervals, 2 h 6 min checked video, 2 h measured active practice, and a 203 h 14 min provisional estimate. The live accounting check passes because active practice remains below checked video. The next chronological pending window is violin 1 at *810-900.

2026-05-13 09:25 PM EDT scan and benchmark update Queue-only owner sync consumed the next two chronological violin 1 windows, *810-900 and *900-990. Both were classified as violin_positive. The follow-up active scan reports 92 sample results, 82 active-violin samples, 10 checked non-violin samples, 257 active sub-intervals, 2 h 10 min checked video, 2 h 3 min measured active practice, and a 201 h 7 min provisional archive estimate. The next chronological pending window is violin 1 at *990-1080. The same pass stored three real user-reviewed score-match failures as benchmark corrections: A4 displayed against a B-position score crop, A4 displayed against a G4 crop, and A5 displayed against a G5 crop. These benchmark fixtures moved the implementation tracker from 29 percent to 31 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-13 10:05 PM EDT scan and exact-percent update Queue-only owner sync consumed violin 1 windows *990-1080 and *1080-1170. Both were classified as violin_positive. The

follow-up active scan reports 94 sample results, 84 active-violin samples, 10 checked non-violin samples, 269 active sub-intervals, 2 h 13 min checked video, 2 h 6 min measured active practice, and a 201 h 24 min provisional archive estimate. The next chronological pending window is violin 1 at *1170-1260. The rounded implementation label remains 31 percent, but exact weighted progress moved to 31.13 percent and is now displayed so small coverage gains are visible. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:14 PM EDT scan and precision update Queue-only owner sync consumed violin 1 windows *1170-1260 and *1260-1350. The first was classified as `violin_positive`; the second was classified as `not_violin_or_unclear`, which adds checked non-playing coverage without adding practice time. The follow-up active scan reports 96 sample results, 85 active-violin samples, 11 checked non-violin samples, 283 active sub-intervals, 2 h 16 min checked video, 2 h 7 min measured active practice, and a 199 h 19 min provisional archive estimate. The next chronological pending window is violin 1 at *1350-1440. The implementation tracker now computes exact weighted progress from unrounded gate points, so the live label should read 31.128 percent after deployment. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:38 PM EDT scan update Queue-only owner sync consumed violin 1 windows *1350-1440 and *1440-1530. Both were classified as `violin_positive`. The follow-up active scan selected both samples, recorded 11 new active sub-intervals, and reports 98 sample results, 87 active-violin samples, 11 checked non-violin samples, 294 active sub-intervals, 2 h 19 min checked video, 2 h 10 min measured active practice, and a 199 h 37 min provisional archive estimate. The next chronological pending window is violin 1 at *1530-1620. Exact weighted implementation progress is now 31.131 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:55 PM EDT scan update Queue-only owner sync consumed violin 1 windows *1530-1620 and *1620-1710. The first was classified as `not_violin_or_unclear`; the second was classified as `violin_positive`. The follow-up active scan selected both samples, recorded 6 new active sub-intervals, and reports 100 sample results, 88 active-violin samples, 12 checked non-violin samples, 300 active sub-intervals, 2 h 22 min checked video, 2 h 12 min measured active practice, and a 197 h 40 min provisional archive estimate. The next chronological pending window is violin 1 at *1710-1800. Exact weighted implementation progress is now 31.134 percent. Exact score windows

and accepted long phrases remain at 0.

2026-05-14 12:09 AM EDT scan update Queue-only owner sync consumed violin 1 windows *1710–1800 and *1800–1890. Both were classified as violin_positive. The follow-up active scan selected both samples, recorded 13 new active sub-intervals, and reports 102 sample results, 90 active-violin samples, 12 checked non-violin samples, 313 active sub-intervals, 2 h 25 min checked video, 2 h 15 min measured active practice, and a 197 h 59 min provisional archive estimate. The next chronological pending window is violin 1 at *1890–1980. Exact weighted implementation progress is now 31.136 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 12:35 AM EDT scan update Queue-only owner sync consumed violin 1 windows *1890–1980 and *1980–2070. Both were classified as violin_positive. The follow-up active scan selected both samples, recorded 9 new active sub-intervals, and reports 104 sample results, 92 active-violin samples, 12 checked non-violin samples, 322 active sub-intervals, 2 h 28 min checked video, 2 h 18 min measured active practice, and a 198 h 18 min provisional archive estimate. The next chronological pending window is violin 1 at *2070–2160. Exact weighted implementation progress is now 31.139 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 03:00 AM EDT scan update Queue-only owner sync consumed violin 1 windows *2070–2160 and *2160–2250. Both were classified as violin_positive. The follow-up active scan folded both samples into the checked ledger and reports 106 sample results, 94 active-violin samples, 12 checked non-violin samples, 331 active sub-intervals, 2 h 31 min checked video, 2 h 21 min measured active practice, and a 198 h 36 min provisional archive estimate. The next chronological pending window is violin 1 at *2250–2340. Exact weighted implementation progress is now 31.142 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 03:23 AM EDT one-measure accelerator update Curtis no longer hardcodes accepted long phrases to zero. The tracker now derives accepted long-phrase count from daily match groups that pass all of the following gates: score location verified, exact score-location status, at least a five-note phrase run, local playable media, and not merely a single-pitch anchor. Regression tests now prove that a verified one-measure symbolic-score phrase with media increments the long-phrase count, while single-note anchors, unverified reference matches, and matches without media do not. This is

implementation progress toward a faster path, not solved musical evidence yet: the live percent remains 31.142 percent until Curtis has a verified symbolic score measure or more scanned archive coverage. The fastest route to visible musical progress is now to verify one Scherzo-Tarantelle symbolic score measure and run the existing phrase matcher across the hidden detected note series already stored for 2026-05-03.

2026-05-14 04:32 AM EDT source-score acceleration update Curtis now carries the Wieniawski Scherzo-Tarantelle solo score PDF as a local source asset rather than depending on weak scanned crops or a network fetch during matching. The score-asset layer reports whether a local source PDF is present, the daily score-reference audit counts local source PDFs separately from symbolic notes, and the 100-point tracker awards source-truth progress for a local source score without counting it as an accepted phrase. Regression tests verify that this raises the score-source readiness state while keeping accepted long phrases, exact score windows, and symbolic score notes at 0. This is the necessary acceleration layer for long-phrase work: the next step is no longer finding a score image, but converting one local source-score measure into verified symbolic notes and running the existing phrase matcher over stored detected note series.

2026-05-14 05:23 AM EDT live symbolic-measure update The deployed Curtis API reported 43.642 percent exact progress, 1 provisional measure match, 1 provisional score-aligned window, 0 accepted long phrases, 1 local score source, and 8 symbolic score notes counted across the two Wieniawski practice-day targets. This state is now superseded: the later source-truth correction rejected the provisional D A# G D source-score claim because the local IMSLP solo part did not support that score-note sequence. The useful implementation gain from this pass was the source-symbolic note gate and score/audio/video/transcription lane, not the specific provisional score phrase.

2026-05-14 05:40 AM EDT source-target accounting update The deployed Curtis API reported 45.142 percent exact progress after the tracker began counting configured source score targets from the reference-training state instead of only local PDF files. That accounting layer remains useful, but the musical evidence count from this pre-correction state is superseded by the 07:52 AM source-truth correction. This makes the progress meter more honest and removes a hidden undercount, but it does not solve phrase transcription; the next implementation step is extending corrected accepted symbolic measure gates into longer matched note runs.

2026-05-14 06:52 AM EDT phrase-candidate ranking update The deployed Curtis API reported 47.642 percent exact progress after the matcher stopped returning the first scan-order matches and instead ranked all detected-series matches by exact score status, score-derived status, distinct pitch-class content, and run length. That state exposed 12 pitch-sequence groups, 1 phrase candidate, 1 provisional measure match, 1 provisional score-aligned window, and 0 accepted long phrases. The highest-ranked pending phrase candidate was D D D# D D A# G. This candidate was not accepted score evidence; it was a focused source-score verification target. The later strict evidence gate supersedes this provisional count.

2026-05-14 07:08 AM EDT tracker visibility update Curtis now returns the highest-ranked pending phrase target as structured tracker state and displays it in the implementation card detail as **pending: D D D# D D A# G**. The visible interface therefore shows not only that one phrase candidate exists, but also which detected pitch-class sequence should be source-score verified next. This does not add accepted musical evidence or change the 47.642 percent full-solve score; it removes a product friction point by making the next verification target visible without presenting it as a solved score match.

2026-05-14 07:52 AM EDT source-truth correction update The Wieniawski symbolic source opening was corrected against the local IMSLP solo-violin score. The old accepted D A# G D match was demoted because the source phrase is D C Bb D C Bb; it can remain only as an unverified reference-sequence candidate, not accepted score evidence. The deployed provisional groups matched detected pitch classes A# D C A# to source score pitch classes A# D C A#, displayed a source-score crop for that phrase, and retained generated notation only as fallback. The tracker then reported 48.642 percent, 2 provisional measure matches, 2 provisional score-aligned windows, 12 pitch-sequence groups, 1 pending phrase candidate, 3 score targets, 1 local score PDF, 6 unique symbolic score notes, 1 source snippet candidate, and 0 accepted long phrases. The later strict evidence gate supersedes these provisional score counts.

2026-05-14 08:29 AM EDT first source-backed phrase update The Scherzo-Tarantelle source excerpt was extended by one verified score note, from D C Bb D C Bb to D C Bb D C Bb D, using a widened crop from the local IMSLP solo-violin PDF rather than the annotated teaching image. That seven-note source phrase let the stored 2026-05-03 detected series match the contiguous five-note run A# D C A# D against source score notes Bb4 D5 C5 Bb4 D5. The deployed API then reported

55.642 percent exact progress, 1 provisional long-phrase candidate, 2 provisional measure matches, 2 provisional score-aligned windows, 12 pitch-sequence groups, 7 unique symbolic score notes, and 1 source snippet candidate. The useful gain was the source-backed path; the later visual review rejected the crop as accepted score evidence, so the strict current state is again 0 accepted long phrases.

2026-05-14 01:50 PM EDT score-coordinate heat-map update Provisional symbolic score phrase matches were connected to score-coordinate heat-map fragments. The 2026-05-03 Scherzo-Tarantelle phrase A# D C A# D was mapped to mm. 2-4 with source score notes Bb4 D5 C5 Bb4 D5, local video, local audio, and an IMSLP score crop. The deployed API reported 58.136 percent exact progress, 1 provisional long-phrase candidate, 2 provisional measure matches, 2 provisional score-aligned windows, 2 provisional score-coordinate heat-map fragments, 12 pitch-sequence groups, 7 unique symbolic score notes, and 1 source snippet candidate. This was a heat-map and evidence-organization advance; the later strict evidence gate blocks those fragments from counting as accepted evidence.

2026-05-14 02:15 PM EDT source-verification target update Curtis now separates the next longer note-run target from accepted score evidence. The tracker exposes one source-verification target, D D D# D D A# G, only because it is a seven-note run from local playable practice media with enough pitch diversity to be worth checking against the score. It remains reference-audio evidence, not accepted score evidence. The source target appears in the tracker and API so the next pass can verify exact local IMSLP score notes and coordinates instead of slowly searching from scratch or accidentally promoting a wrong crop. The deployed tracker state then included provisional score artifacts, but the current strict evidence gate counts 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, and 0 verified source snippets until source review passes.

2026-05-14 02:51 PM EDT source-target triage update Curtis now checks four-note-or-longer source targets against the verified symbolic score map before promotion. The current top target D C G A was compared against the verified source excerpt D C A# D C A# D. The best contiguous overlap is two notes, so the target is recorded as `source_score_sequence_not_found` and remains reference-audio evidence only. The tracker then reported 58.633 percent exact progress, 3 source-target score checks, 0 verified source targets, and the next action: extend the verified IMSLP symbolic score map before promoting this sequence. This improves the long-phrase pipeline by making measure-sized

candidates fail fast and explicitly instead of appearing as misleading score matches.

2026-05-14 03:49 PM EDT score-map workbench update Curtis now has an explicit source-map acceleration layer for the local IMSLP Wieniawski solo part. A deterministic extraction tool renders page 2 of the local score PDF, detects staff-line groups, removes staff lines, extracts connected score glyph candidates, and writes a JSON verification queue. The current queue contains 779 unaccepted score glyph candidates across 7 staves. These candidates are not pitch labels, not accepted score notes, and not displayable score evidence. They exist to make the next work cycle faster: verify candidate glyphs against the local PDF, convert only verified notes into MusicXML, then run source-target matching against verified symbolic notes. The tracker now reports 59.633 percent exact progress after awarding score-truth credit for the verification queue while preserving 0 verified source targets for the current D C G A source target.

2026-05-14 04:24 PM EDT note-hypothesis acceleration update The source-map workbench now filters the local IMSLP glyph queue into likely notehead candidates and assigns staff-position pitch hypotheses using the G-minor key signature. The current queue contains 172 unaccepted score-note hypotheses across 7 staves in addition to the 779 raw glyph candidates. These hypotheses are explicitly not accepted score evidence: they use staff position only, may include false positives, and cannot create a score snippet, pitch label, or match card until source-reviewed into MusicXML. The implementation value is speed. The next pass can review a note-labeled queue instead of starting from anonymous glyph boxes, which should make extending the Scherzo-Tarantelle symbolic map to full measures and longer phrases much faster. The tracker now reports 61.133 percent exact progress after awarding limited score-truth credit for the note-hypothesis queue while preserving 0 verified source targets for the current D C G A source target.

2026-05-14 05:31 PM EDT staff-review packet update The score-map detector was changed from a single-threshold staff finder to a multi-threshold merged staff finder after review showed that the previous queue skipped a source staff. Page-2 coverage now reports 951 unaccepted score glyph candidates and 214 unaccepted notehead hypotheses across 8 staves. The same pass writes 8 staff-level review-packet images under the local score assets folder, with each packet showing the scanned IMSLP staff plus numbered hypothesis boxes. The packets are not accepted score evidence and do not change any match card by themselves; they exist to make the next MusicXML pass faster and less error-prone

by turning source review into a bounded staff-by-staff checklist. The deployed tracker now reports 64.135 percent exact progress after awarding score-truth credit for complete source-review packet generation while preserving 0 verified source targets for the current D D D# D D A# G source target, whose verified symbolic-score overlap is still only 1/7.

2026-05-14 06:15 PM EDT score/transcription visual-agreement update The public 2026-05-03 match had a real failure: the score panel showed a broad opening source crop even though the claimed transcription phrase was the narrower five-note slice Bb4 D5 C5 Bb4 D5. Curtis then added a separate score-display gate: a score panel could render an actual source crop only when the crop's reference start and end exactly matched the phrase, and broad covering crops were withheld. The symbolic matcher also began requiring MIDI-level equality between detected notes and parsed score notes, not only pitch-class similarity, before it could create a score-visual lock. The deployed tracker reported 64.878 percent exact progress and provisional score locks, but later visual review showed the crop still was not trustworthy enough; the current strict gate counts this phrase as pending, not accepted.

2026-05-14 07:04 PM EDT actual-source score-gate update The previous visual-agreement fix still left one unacceptable loophole: a generated notation SVG could appear in the accepted score panel even though Alan asked for the actual piece. Curtis therefore required a score panel to use a non-generated source-score crop for the same reference range. The five-note 2026-05-03 Scherzo-Tarantelle match gained an exact-range source crop for Bb4 D5 C5 Bb4 D5; generated notation remained only as the transcription rendering. That deployment expected provisional actual-source score progress, but the later musician review rejected the crop and the current strict evidence gate blocks it from creating an accepted measure, long-phrase count, exact score-aligned window, score-coordinate heat-map fragment, actual-source score lock, or verified source snippet.

2026-05-14 09:45 PM EDT score-spelling visual-context update The latest reported mismatch exposed another gate that had been too weak: a transcription could be score-spelled as Bb4 D5 C5 Bb4 D5 in data while the rendered notation panel omitted the source key signature, making the first and fourth notes look like unqualified natural-position B notes to a reader. Curtis now carries the source key signature into score-matched transcription rendering and requires both `scoreVisualRangeAgreement` and `scoreSpellingAgreement` before an actual-source score crop can count as accepted score

evidence. The Scherzo-Tarantelle match remains a five-note fragment, but the visible transcription and visible score must now agree in musical spelling context rather than only MIDI or pitch class. This is a stricter acceptance gate, not a new long-phrase solve.

2026-05-15 12:29 PM EDT exact-note truth-gate update The reported failure mode was not only visual range: a score match could still be treated as acceptable if the pitch class agreed while the written note/register did not. The truth workbench now stores MIDI sequences for detected, accepted, and score notes; accepted score-truth items reject audio-score mismatches at exact note-and-octave level; and score-truth items cannot be accepted with score notes that omit octave/register. Single-note score anchors also require an explicit exact-note verification flag before the UI can render them. Regression tests now cover accepted A4–D5 truth, A4-versus-A5 rejection, score truth without octave rejection, exact pitch-anchor display, and visual-only pitch-anchor rejection. This does not create a new accepted phrase; it prevents Curtis from ever calling a pitch-class-only or visually guessed crop solved evidence.

2026-05-14 10:52 PM EDT visual-crop demotion update Alan's follow-up review showed that the notes visible in the 2026-05-03 source crop still did not match the transcription. Curtis now rejects the Scherzo-Tarantelle source crops as review material unless the visible noteheads, register, written spelling, and exact range are independently verified against the transcription. The five-note 5/3 phrase remains a symbolic phrase candidate, but it no longer creates an accepted score panel, accepted measure match, accepted long phrase, exact score-aligned window, score-coordinate heat-map fragment, score-visual lock, actual-source score lock, or verified source snippet. This is a regression lock: wrong score evidence is removed rather than cosmetically relabeled.

2026-05-15 12:45 AM EDT truth-workbench update Curtis now has a persistent truth-workbench API and compact implementation-status surface for accepted, pending, and rejected audio-score-transcription evidence. A truth item can store the source clip, practice day, piece, detected notes, accepted notes, score notes, score source, score location, score image, and gate state. Accepted truth items are rejected at write time if the accepted audio note sequence and score note sequence disagree. The backend and frontend score gates also now require explicit visible-note-sequence verification before a score panel can count as a score-visual lock, actual-source score lock, accepted measure match, or accepted long phrase. This prevents the exact failure Alan found: a visually wrong score crop cannot count just because the metadata says the pitch sequence should match. It still has 0 accepted long

phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, 0 score-visual locks, 0 actual-source score locks, and 0 accepted truth clips.

2026-05-15 03:20 PM EDT exact-source phrase attempt Curtis temporarily promoted one actual-source Scherzo-Tarantelle phrase crop, Bb4 D5 C5 Bb4 D5 from the local IMSLP page-2 solo-violin scan, aligned to the symbolic score range mm. 2–4. The attempt boxed five matched noteheads and stored source page pixel coordinates and exact visible note-plus-octave sequence. This pass was useful because it proved the crop extraction path could render actual score material, but it was later rejected by visual review because the score notes and boxes were not trustworthy enough to count as accepted evidence.

2026-05-15 03:40 PM EDT superseded deployment verification The deployed Curtis site served cache version 20260515-exact-source-phrase on deployment 4fb78ce1. That deployment still showed one score group for 2026-05-03, with detected notes A#4 D5 C5 A#4 D5, score-spelled transcription Bb4 D5 C5 Bb4 D5, source score notes Bb4 D5 C5 Bb4 D5, the exact-source crop asset, local video sample Njh8_zq9_DM-9465, and local audio from 6.618–7.674 s inside that sample. This state is now superseded by the 05:00 PM EDT strict evidence-gate pass because the displayed score crop did not survive musician visual review. The live practice-time surface reported 234 h 57 min uploaded archive video, 2 h 28 min checked video, 2 h 21 min measured active violin, 223 h 5 min estimated total active practice, and 232 h 28 min unchecked video. The estimate remains provisional until the full chronological active-practice scan covers every archive window.

2026-05-15 05:00 PM EDT strict evidence-gate update Curtis now rejects the 2026-05-03 Scherzo-Tarantelle source crop as accepted evidence unless the full truth chain passes: source note boxes centered on the visible noteheads, visible score notes matching the claimed written notes and octaves, transcription notes matching the paired audio, transcription notes matching the score notes, and an accepted truth-workbench item tying the audio, transcription, and score together. Backend, frontend, tracker, and tests now require those fields before a score snippet, accepted measure match, exact score-aligned window, actual-source score lock, or long-phrase count can appear. The deployed tracker reports 43.126 percent exact weighted progress, 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 actual-source score locks, and 0 verified source snippets until a crop survives this stricter review.

2026-05-15 04:01 PM EDT practice-time API unification The daily-record API now consumes the same active-practice coverage ledger used by `/api/curtis/active-practice-coverage` before publishing top-level and per-day practice totals. This removes the stale split where one public route could show older transcription-derived totals while the coverage ledger showed 2 h 28 min checked video, 2 h 21 min measured active violin, 223 h 5 min estimated total active practice, and 232 h 28 min unchecked video. A regression test now verifies that daily records inherit canonical checked, active, estimate, unmeasured, and per-day scan totals from the coverage ledger.

5.2 Definition of solved

Long-phrase transcription is solved for Curtis only when the system can take Alan's long-form practice archive and produce evidence groups that a musician can trust without manually checking every note. The complete goal is not merely a model that prints plausible notation. The solved system must satisfy all of the following conditions.

1. **Full archive accounting.** Every practice video from the strict practice ledger is assigned to the correct practice day, has exact uploaded duration, and has a measured or explicitly pending active-practice state.
2. **Active violin time.** Total practice time means intervals where Alan is actually playing violin. Talking, tuning, setup, walking, camera adjustment, silence, scrolling, and other non-playing intervals are excluded.
3. **Local evidence playback.** Every accepted transcription snippet has a playable local video clip and audio clip in the Curtis site. Accepted evidence does not rely on a YouTube embed.
4. **Accurate note events.** The transcribed notes match the paired audio, including repeated notes, fast scalar runs, arpeggios, shifts, string crossings, rests, and obvious restarts. A fast passage must not collapse into a false repeated pitch stream such as D D D D.
5. **Rhythm and timing.** The system estimates onsets, durations, rests, tempo changes, slow-practice stretches, restarts, and repeated attempts. When rhythm is uncertain, the notation must mark uncertainty rather than invent a clean answer.
6. **Professional notation.** Displayed notation uses a real notation engine or a notation-quality rendering path. Treble clef placement, staff scaling, note placement, stems, beams, rests, spacing,

- key signature, and measure layout must look like real sheet music, not a hand-drawn mockup. 715
7. **Score-source truth.** A score snippet can appear only when the score-side notes are source-confirmed. Scanned images without a verified note map are not enough. A displayed A must point to an actual A in the score. 716
717
718
 8. **Score alignment.** For known repertoire, the played note sequence must align to a specific location in the score. The match can ignore rhythm during the first pitch-class search, but accepted score display needs exact score-note identity and a stable location. 719
720
721
 9. **Score-free exercise handling.** If Alan invents an exercise or practices a technique pattern that is not in the score, Curtis should still transcribe and group the pattern. It should not force the pattern into Scherzo-Tarantelle, Haydn, or another piece. 722
723
724
 10. **Repeat grouping.** Repeated attempts of the same passage are grouped as attempts, loops, restarts, or slow/faster passes. The UI should not dump the same measures repeatedly. 725
726
 11. **Heat maps.** Practice density, repetition density, problem density, and improvement layers appear on actual score coordinates when a score match exists, or on the generated pattern/transcription when the material is score-free. 727
728
729
 12. **Curtis-level observations.** Observations are specific and evidence-tied: what happened, where it happened, how often it happened, whether it improved, and why the issue blocks Curtis-level readiness. Generic advice is not accepted. 730
731
732
 13. **Compact interface.** Expanded daily entries stay compact. The useful unit is the grouped score–transcription–audio–video pair, not a large dashboard or a long explanatory panel. 733
734
 14. **Regression protection.** Wrong score-note boxes, failed transcriptions, unmatched notation, stale piece labels, and non-playing practice-time credit cannot silently reappear after a later change. 735
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5.3 Acceptance gates 738

The following gates define whether progress is real. If a change fails one of these gates, it is not counted as solved, even if the UI looks better. 739
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Archive gate.

The strict practice ledger must list every video from violin 1 forward that is either violin-numbered or date-titled as practice. Same-day videos must merge into one daily record. Each record stores source video id, title, uploaded date, inferred practice date, uploaded duration, and processing coverage.

Practice-time gate.

Active violin time is the sum of detected playing intervals, not the uploaded duration. The API must expose measured active time, checked video time, unmeasured video time, and estimate status separately. The UI label should use total practice time for active violin time and uploaded video for total source duration.

Evidence-media gate.

A transcription line is accepted only if the exact audio and video window are available locally, playable in the same Curtis page, and time-linked to the source record.

Note gate.

A displayed note must agree with the paired audio. The note label, octave where claimed, and staff position must be internally consistent. If the audio says A4, the rendered staff note must be A4, not F, G, B, or a visually adjacent note.

Score gate.

An accepted score snippet must come from the actual piece source: a verified source-score crop, verified MusicXML engraving export, or manually verified note-coordinate map for the same reference range. Generated notation may show the transcription, but it cannot appear as the accepted score snippet.

Single-note gate.

A one-note score display is allowed only as a calibration anchor, and only if both sides explicitly show the same pitch name and the score-side coordinate is verified. It must not be used as proof of a long passage.

Phrase gate.

A real phrase match requires a contiguous pitch-class sequence in the score or pattern source.

Rhythm can be ignored for early matching, but accepted phrase display must keep the exact played sequence, the exact score sequence, and any omitted or uncertain notes.

Notation gate.

Hand-built notation SVG is not the long-term display path. The production path must use a professional renderer or renderer-equivalent layout with verified clef, key, staff, note, rest, spacing, and responsive behavior.

Exercise gate.

If no score match exists and the material forms a repeated exercise, the system must label it as an exercise or pattern, preserve the generated transcription, and avoid adding a fake repertoire entry.

Observation gate.

A Curtis-level observation must cite the clip, transcription event, score measure or pattern segment, issue type, repeat count, direction of change, and readiness consequence.

UI gate.

The day row stays chronological, the expanded state is compact, and only grouped evidence appears by default. Raw untrusted transcription stays hidden or goes to the transcription-run PDF with clear status.

Paper gate.

Any meaningful progress changes this section, the system-state table, and the limitation text in the same work cycle.

5.4 Required source material

The implementation cannot be completed by code alone. It needs a source bundle that makes the audio, score, and corrections checkable. The Wieniawski Scherzo-Tarantelle solo PDF is now present locally as the first source-score asset, and its first verified symbolic map covers the seven-note opening phrase only.

1. **Raw or locally captured practice media.** Curtis needs stable local access to audio/video windows from the YouTube archive or owner-provided files. Metadata alone cannot produce practice time, clips, or transcription.

2. **Source-confirmed piece names by day.** Alan can supply piece names when automatic detection is uncertain. These labels should seed matching but never become score evidence by themselves.
3. **Symbolic scores for known repertoire.** For each piece, Curtis needs MusicXML, MEI, a verified MuseScore export, or an equivalent symbolic note map with measure numbers, pitches, durations, voices, and staff positions.
4. **Score images tied to symbolic coordinates.** To show actual score snippets, the image/PDF rendering must be linked to the same symbolic note ids or verified coordinate boxes.
5. **Manual corrections.** Alan or a reviewer must be able to correct a note, octave, onset, rhythm, score location, or false match. These corrections become training and regression data.
6. **Calibration recordings.** Labeled scales, arpeggios, open strings, repeated notes, shifts, slow exercises, and fast exercises help tune pitch detection before attempting long phrases.
7. **Gold benchmark clips.** A small set of manually checked audio–transcription–score pairs is required before the app can claim improvement. The benchmark must include easy anchors, repeated notes, fast runs, arpeggios, and real Scherzo-Tarantelle fragments.

5.5 Implementation plan

Phase 0: Make the paper the tracker

Status: complete in this revision. The first required step is to make the plan durable in the Curtis paper PDF. The paper now records the definition of solved, the current baseline, the gates, the phase plan, and the update protocol. Future changes are not complete unless the paper is checked or updated.

Phase 1: Build the canonical evidence ledger

The backend needs a stricter data model for practice evidence. The current modules already include the daily-record builder, media sampler, transcription engine, symbolic-score parser, score-asset layer, reference corpus, and correction layer. The next version should make the following entities explicit:

- **source_video:** immutable source id, title, URL, uploaded time, duration, practice-date basis, and ledger inclusion basis.

- **practice_day**: date, list of source videos, uploaded duration, active-practice coverage, piece labels, and status. 821 822
- **active_interval**: source id, start, end, confidence, detector version, reason for inclusion, and reason for exclusion when withheld. 823 824
- **media_clip**: local video path, local audio path, waveform path, source interval, duration, and playable route. 825 826
- **note_event**: onset, offset, pitch class, octave, cents deviation, confidence, detector sources, uncertainty reason, and correction state. 827 828
- **score_note**: piece id, movement, measure, voice, pitch class, octave, duration, note id, and rendered coordinates. 829 830
- **match_candidate**: played note ids, score note ids or exercise-pattern ids, match score, mismatch list, and display eligibility. 831 832
- **accepted_evidence**: only the groups that pass the gates; these are the only groups the main UI can show. 833 834
- **rejected_evidence**: wrong notes, wrong score boxes, false labels, failed transcriptions, and model guesses that must not reappear. 835 836

The acceptance condition for this phase is that a wrong A-to-B or A-to-G score crop can be stored as rejected evidence and cannot be selected by the display builder again. 837 838

Phase 2: Complete the full active-practice scan 839

The practice-time problem is separate from transcription. Curtis should first scan the full 234 h 57 min uploaded practice ledger for violin-playing intervals with a low-cost pass. The pass should use audio energy, violin-range voiced pitch, onset density, spectral shape, and visual fallback only where necessary. It should not require full transcription. 840 841 842 843

The fast pass should walk every practice video in chronological order and store approximate active intervals. A second exact pass should refine boundaries around detected regions. The UI should show three numbers for every day: uploaded video time, checked video time, and total practice time. While 844 845 846

scanning is partial, the estimate stays labeled as estimated. When every video has been scanned, the estimate should be replaced by measured total practice time.

Acceptance for this phase:

- every strict-ledger video has a coverage record;
- every day has uploaded duration and active-practice status;
- non-playing windows are excluded;
- active-practice total can be recomputed from stored intervals;
- the paper table is updated with checked duration, active duration, unmeasured duration, and estimate status.

Phase 3: Generate stable local evidence clips

Every accepted interval and every candidate transcription window needs local media. Curtis should generate short MP4 clips and audio clips with stable IDs derived from source video id and time range. The browser should play the local clip and audio without YouTube embedding. The audio used for transcription must be the same audio the user can play.

Acceptance for this phase:

- each accepted transcription event links to local video and local audio;
- the site can play the clip in the same window;
- source time, local file, and rendered transcription agree;
- dead local media routes fail the evidence gate rather than rendering a broken accepted group.

Phase 4: Build the verified score library

The current weak link is score-side truth. Curtis must stop relying on guessed scanned-score boxes. For Wieniawski Scherzo-Tarantelle and any future piece, the score library needs symbolic notes and rendered coordinates.

The best path is:

1. acquire a legal public-domain score image or PDF; 871
2. obtain or create a symbolic MusicXML/MEI/MuseScore representation; 872
3. render the symbolic score to image or SVG with stable note ids; 873
4. map each rendered note to pitch, octave, measure, and coordinates; 874
5. manually verify a representative set of anchors, especially A, D, G, B, fast runs, ledger lines, 875
accidentals, and octave-displaced passages; 876
6. reject any score source where the note map cannot distinguish adjacent notes reliably. 877

Acceptance for this phase: 878

- a score-side A in the UI is generated from an actual score-side A; 879
- the rendered crop includes enough context to be legible; 880
- the crop does not cut off clefs, notes, beams, stems, or measure context; 881
- single-note anchors are visually verified before display; 882
- phrase snippets use the exact matched symbolic notes, not a guessed image region. 883

Phase 5: Replace fragile note extraction with a multi-stage transcription engine 884

The note detector needs to work on violin practice, not only on clean isolated notes. The engine should 885
process active intervals in stages: 886

1. normalize audio and remove long silence; 887
2. isolate violin-dominant energy where possible; 888
3. detect onsets and repeated attacks before pitch smoothing; 889
4. run multiple pitch estimators over the same window; 890
5. separate stable pitch, glissando/shift, vibrato, noise, and uncertain frames; 891
6. group frames into note events by onset, pitch stability, and duration; 892

7. preserve fast pitch changes instead of smoothing them into one repeated pitch;
8. estimate rests and pauses between attempts;
9. detect double stops or chords as multi-note events when monophonic assumptions fail;
10. attach confidence and uncertainty to every event;
11. store rejected alternatives so repeated failures can be debugged.

Acceptance for this phase:

- open strings and simple scales transcribe correctly;
- repeated notes remain repeated notes with distinct onsets;
- arpeggios become changing pitch sequences, not a repeated-note collapse;
- obvious fast runs preserve the contour and most pitch classes;
- uncertain events are labeled uncertain rather than shown as clean notation;
- every displayed note can be played from its paired audio.

Phase 6: Make transcription score-aware

For known repertoire, Curtis should not transcribe in isolation and then guess the piece later. The user-supplied or detected piece name should provide a score prior. The aligner should search the score for pitch-class sequences that match the audio-derived note stream, allowing restarts, skipped notes, slow practice, repeated loops, tempo changes, and rhythm mismatch during early search.

The alignment should use multiple layers:

- pitch-class subsequence search for easy anchors;
- interval-contour search when absolute pitch is uncertain;
- dynamic time warping for variable tempo;
- Viterbi-style path selection through the symbolic score for longer passages;
- restart and loop detection when the path jumps backward;

- score-free fallback when no score path is plausible. 916

Acceptance for this phase: 917

- one-note anchors pass only when the score coordinate is verified; 918
- three-to-five-note sequences can be matched to exact score notes; 919
- longer phrases show played notes beside matched score notes; 920
- mismatches are visible rather than hidden; 921
- rhythm mismatch can be tolerated during matching but not during final notation labeling; 922
- accepted score matches survive a musician spot-check. 923

Phase 7: Support score-free technique exercises 924

Alan may invent an exercise during practice. Curtis must treat this as first-class evidence. If the note stream is coherent but does not match the named score, the system should cluster repeated patterns, render generated notation, and label the material as a technique pattern or exercise. 925
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Acceptance for this phase: 928

- repeated made-up patterns group together; 929
- the generated transcription is linked to audio/video; 930
- no fake score snippet appears; 931
- the repertoire entry records technique-pattern evidence separately from piece evidence; 932
- heat maps can show pattern density without pretending there is a score location. 933

Phase 8: Add manual correction and training 934

Fully automatic transcription will not become reliable without corrections. Curtis needs a correction UI that allows Alan to say: this note is A, this staff note is wrong, this score box points to B, this clip is non-playing, this rhythm is wrong, this is Scherzo-Tarantelle, this is a made-up exercise, or this score match is false. 935
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Every correction should become data: 939

- corrected note events become gold labels;
- rejected matches become negative examples;
- accepted matches become benchmark fixtures;
- corrected piece names update source labels without unlocking score snippets by themselves;
- corrections invalidate stale derived evidence.

Acceptance for this phase:

- Alan can fix a wrong note or score box without editing code;
- the corrected example appears in the benchmark set;
- the same failure cannot reappear after rerendering;
- the PDF tracker records the correction count and benchmark count.

Phase 9: Build the benchmark suite

Curtis needs objective tests before claiming improvement. The benchmark should include:

- open-string clips;
- G, D, A, and E major/minor scales where available;
- arpeggios;
- repeated single-note attacks;
- fast scalar runs;
- shifts with sliding or unstable pitch;
- Scherzo-Tarantelle fragments with verified score locations;
- Haydn fragments with verified score locations;
- score-free exercises;
- non-playing windows that must not count as practice time.

Metrics should include note precision, note recall, pitch-class accuracy, octave accuracy when claimed, onset error, duration error, false-note rate, repeated-note recall, phrase-match precision, phrase-match recall, score-coordinate accuracy, false accepted score match count, active-practice precision, active-practice recall, and UI evidence completeness.

Acceptance for this phase:

- tests fail on the known A-to-wrong-score-note examples;
- tests fail on repeated-pitch collapse;
- tests fail when accepted notation lacks paired audio/video;
- tests fail when practice time includes non-playing footage;
- tests pass only when the displayed evidence is actually trustworthy.

Phase 10: Use professional notation rendering

The current custom notation path is only a temporary evidence display. The solved system should render notation from MusicXML, MEI, or another structured notation representation through a professional renderer or renderer-equivalent layout. The renderer must handle treble clef placement, staff scale, ledger lines, rests, stems, beams, ties, slurs where known, key signatures, accidentals, note spacing, measure widths, and responsive crop behavior.

Acceptance for this phase:

- treble clef is correctly centered around the G-line spiral;
- the staff note for A4 appears in the A4 position;
- key signature and accidentals follow the chosen notation policy;
- snippets look publication-quality at desktop and mobile widths;
- the same notation source can export to the transcription-run PDF.

Phase 11: Build heat maps from matched coordinates

Heat maps should be computed from actual practice behavior. For score-matched passages, each matched score note or measure receives density values. For score-free exercises, each pattern segment

987 receives density values.

988 The layers are:

- 989 • practice density: active time spent on the passage;
- 990 • repetition density: number of attempts or loops;
- 991 • problem density: restarts, stops, slowdowns, unstable notes, and failed attempts;
- 992 • improvement: later attempts cleaner than earlier attempts.

993 Acceptance for this phase:

- 994 • a heat-map mark can be traced to clips and note events;
- 995 • a score-free exercise can still have a pattern heat map;
- 996 • heat maps do not appear as decorative overlays when no match exists;
- 997 • the day-level heat map and repertoire heat map use the same evidence.

998 **Phase 12: Generate specific Curtis-level observations**

999 Observation generation should happen after transcription and alignment, not before. The observation
1000 engine should compare attempts of the same passage and classify issues as pitch, rhythm, bow, shift,
1001 tone, articulation, tempo, consistency, musical shape, or visible tension.

1002 The output for each main passage should include:

- 1003 • likely passage or exercise pattern;
- 1004 • evidence clip;
- 1005 • transcription snippet;
- 1006 • score snippet when verified;
- 1007 • exact issue;
- 1008 • number of attempts where it occurred;
- 1009 • whether it improved, stayed the same, or got worse;
- 1010 • why it blocks Curtis-level readiness.

Acceptance for this phase:

- observations cite evidence; 1011
- generic advice is absent; 1012
- a day ends with one specific main blocker; 1013
- rejected or uncertain transcription does not feed readiness claims. 1014

Phase 13: Keep the UI compact

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The main page should show the paper card, analyzed practice days, and repertoire. Expanded daily 1017
entries should show one or a small number of grouped evidence pairs, not a full debug dashboard. The 1018
day marker should remain visible on the left so boundaries between days are clear. Videos should be 1019
small enough that a day entry fits on screen when possible. 1020

Acceptance for this phase:

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- one daily group contains score, clip, audio, transcription, and the minimum useful status; 1022
- raw text explanations are removed from the main view; 1023
- chronological order is stable; 1024
- desktop and mobile render without overlap, excessive height, or broken media controls; 1025
- unresolved material is visible as state, not as a fake dashboard. 1026

Phase 14: Automate paper updates

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The paper is now the progress tracker. Future meaningful Curtis changes should update:

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- live state numbers; 1029
- solved gates; 1030
- benchmark metrics; 1031
- accepted score-match count; 1032
- accepted transcription duration; 1033

- 1034 • full archive scan coverage;
- 1035 • remaining blockers;
- 1036 • limitations.

1037 Acceptance for this phase is procedural: after every Curtis change that affects transcription, score
1038 matching, practice time, clips, heat maps, repertoire, observations, or evidence gates, the LaTeX paper
1039 is checked, rebuilt, and the root paper .pdf is refreshed.

1040 5.6 Milestones

1041 5.7 Immediate next implementation

1042 After this PDF tracker, the first concrete implementation target is **full active-practice coverage plus**
1043 **benchmark/correction scaffolding**. This is the first useful code milestone because transcription
1044 cannot be trusted without knowing which source windows contain violin playing, and it cannot improve
1045 reliably without a gold correction loop.

1046 The 2026-05-13 11:57 AM EDT backend update completed the coverage and correction scaffolding
1047 portion:

- 1048 • added a canonical active-practice coverage ledger exposed through the review API and
1049 /api/curtis/active-practice-coverage;
- 1050 • separated uploaded video duration, checked sampled video duration, measured active practice time,
1051 candidate violin windows, unmeasured video time, and estimate state;
- 1052 • counted checked non-playing sampled windows as checked video without adding them to practice
1053 time;
- 1054 • added /api/curtis/evidence-corrections for note, score, match, piece, and active-interval
1055 corrections;
- 1056 • added benchmark fixtures for rejected score-note and match corrections;
- 1057 • added regression tests for checked non-playing windows, wrong score-note benchmark creation,
1058 and rejecting accepted score evidence when the displayed score note conflicts with the observed
1059 note.

1060 The 2026-05-13 12:33 PM EDT backend update completed the active-scan route portion:

Table 4. Milestone plan for moving from current evidence gates to solved long-phrase transcription.

Milestone	Target	Acceptance evidence	Status
M0	Paper tracker	Roadmap is included in the Curtis paper PDF and becomes the update record.	Complete in this revision.
M1	Practice time	All 234 h 57 min of strict-ledger video has active-practice coverage and per-day totals.	Active-scan route deployed; live state now has 331 active intervals from 106 sample results, 2 h 28 min checked video, 2 h 21 min measured active practice, and 250 queued windows; full chronological scan still pending.
M2	Score truth	Scherzo-Tarantelle has verified symbolic notes and score coordinates.	Partially advanced; the local source PDF is present and a seven-note symbolic opening phrase is now parsed from source-backed MusicXML. More measures and score coordinates remain pending.
M3	One-note anchor	A displayed A from audio is matched to a verified score-side A, with no wrong-note crop.	Pending.
M4	Short sequence	Three-to-five played notes align to exact score notes and render with local audio/video.	Advanced; the matcher can now accept source-backed measure fragments and the first five-note source-backed phrase with local media, while longer passages still require the full phrase gate.
M5	Long phrase	A 10–30 s passage transcription matches the paired audio and score location.	Pending.
M6	Repeat grouping	Multiple attempts of the same passage group into loops, restarts, slow attempts, and faster attempts.	Pending.
M7	Heat map	Practice density and problem density appear on verified score coordinates.	Pending.
M8	Repertoire update	Piece entries update from accepted daily evidence and show active time, clips, snippets, and blockers.	Pending.
M9	Readiness observation	Main blocker is generated from accepted evidence, not generic advice.	Pending.
M10	Regression lock	Tests prevent failed notation, wrong score boxes, and non-playing practice-time credit from returning.	Correction benchmark scaffold added; 3 user-reviewed wrong-score-note regressions are stored; truth-workbench write gates reject accepted audio-score note mismatches; visible-note-sequence verification is required for accepted score panels; full benchmark suite pending.

- 1061 • added a persistent `activePracticeScan` state record with active intervals, sample results, pending
1062 windows, and run history;
- 1063 • scans stored local media samples only after the violin-presence gate is positive;
- 1064 • extracts sound-active sub-intervals inside the source window and stores them as active-practice
1065 intervals;
- 1066 • records negative violin-presence samples as checked without adding practice time;
- 1067 • feeds persisted active intervals into per-day totals and the active-practice coverage ledger;
- 1068 • exposes active-scan status and run endpoints so the deployed runtime can perform the pass without
1069 transcription;
- 1070 • queues the next unchecked 90-second strict-ledger windows for owner media capture.

1071 The 2026-05-13 12:59 PM EDT deployed scan update completed the first live run and exposed a
1072 practice-time accounting failure mode: sparse, captured samples can make an archive-wide estimate
1073 look too certain. The implementation now records active-scan status counts, caps active seconds by
1074 checked source coverage, and treats active-scan results as the practice-time source for scanned samples
1075 instead of letting noisy transcription durations override them.

1076 The 2026-05-13 03:26 PM EDT queue update connected the active-scan queue to the owner-media
1077 sync path. The sync path now asks the deployed active-scan run endpoint for pending windows, keeps
1078 the response compact, and places those windows ahead of generic expansion or blind sampling. The
1079 pending queue remains the source of work for M1: each captured window must still be downloaded or
1080 browser-captured, locally classified for violin presence, uploaded, rescanned, and recorded before it
1081 becomes checked practice-time coverage.

1082 The 2026-05-13 03:37 PM EDT owner-sync run proved that the local upload/rescan loop still works
1083 after the queue update. It uploaded one cached violin-positive 5/3 window and the deployed active
1084 scan converted it into one additional active-practice interval. Because cached violin-positive media
1085 is still prioritized ahead of new capture, this run did not consume the first chronological pending
1086 window. The next M1 step should either exhaust the cached positive backlog first or run a queue-only
1087 owner-sync mode so the pending list starting at `violin 1` advances.

1088 The 2026-05-13 04:05 PM EDT update added that queue-only owner-sync mode and used it to consume
1089 the first chronological pending window. The `violin 1 *0-90` window was captured, classified as

non-playing/unclear, uploaded, and folded into the scan results as checked non-violin evidence. This is correct progress because it prevents setup or non-playing footage from being counted as violin practice. It does not advance transcription because no violin-positive musical material was present in that window. The UI now shows the current long-phrase transcription completion percentage and the complete done/open gate list at the bottom of the Curtis page so the remaining work is visible without turning daily entries into a large debug dashboard.

The 2026-05-13 05:44 PM EDT update advanced the same queue from *90-180 through *450-540. Those windows produced active-practice evidence and moved checked coverage to 2 h 3 min. The timeout exposed a resume failure: cached active-scan pending windows could be captured but then skipped on the next queue-only run. The helper now includes matching cached pending-window files in queue-only order, including non-playing cached windows, so an interrupted scan can resume rather than forcing Alan or Codex to manage the cache manually. The coverage ledger also now treats unchecked transcription-only remnants as zero practice time; active practice cannot exceed checked video coverage.

The 2026-05-13 06:01 PM EDT update consumed the next chronological queue window, *540-630. That window was violin-positive and added four active intervals after the active-practice scan. The run succeeded without a timeout at batch size one, and the next pending window became *630-720. This kept the current work focused on M1 coverage rather than score matching or long-phrase notation.

The 2026-05-13 06:26 PM EDT update consumed the next chronological queue window, *630-720. That window was violin-positive and added five active intervals after the active-practice scan. The command wrapper timed out while the owner-sync process was still running, so the process was allowed to finish and then verified from live state instead of starting a duplicate run. The next pending window is now *720-810.

The 2026-05-13 07:55 PM EDT update consumed the next chronological queue window, *720-810. That window was violin-positive and added three active intervals after the active-practice scan. The owner-sync process completed cleanly with a longer timeout, so no duplicate capture or manual cache recovery was needed. The next pending window is now *810-900.

The 2026-05-13 08:41 PM EDT status-surface update made the Curtis page itself carry the implementation state instead of leaving the percentage and plan in the chat thread. The bottom tracker is now labeled as long-phrase transcription implementation status and is driven by the same backend

gate object as the API. The visible panel reports 29 percent completion, 2 h 7 min checked video out of 213 h 23 min, 2 h measured active practice, 0 exact score windows, 0 accepted long phrases, 250 queued windows, and 250 active intervals. It also exposes a compact done list, an open-work list, an eight-phase implementation plan, and a collapsed detailed gate checklist. This is a presentation and tracking improvement, not a transcription breakthrough: score-truth verification, phrase-level note/rhythm extraction, exact score alignment, heat maps, and Curtis-level observations remain open. The 2026-05-13 09:25 PM EDT implementation pass advanced the M1 scan and M10 benchmark loop. The queue-only sync consumed violin 1 windows *810-900 and *900-990; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31 percent implementation completion, 2 h 10 min checked video out of 213 h 23 min, 2 h 3 min measured active practice, 92 sample results, 82 active-violin samples, 10 checked non-violin samples, 257 active intervals, and 250 queued windows. The correction ledger now contains three benchmark fixtures for real wrong-score-note failures: A4/B-crop, A4/G4-crop, and A5/G5-crop. Those cases are progress only because they prevent the same bad score matches from being counted as solved evidence. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 10:05 PM EDT implementation pass continued the M1 scan and fixed a tracker presentation gap. The queue-only sync consumed violin 1 windows *990-1080 and *1080-1170; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31.13 percent exact weighted implementation completion, 2 h 13 min checked video out of 213 h 23 min, 2 h 6 min measured active practice, 94 sample results, 84 active-violin samples, 10 checked non-violin samples, 269 active intervals, and 250 queued windows. The rounded public label still reads 31 percent, but the API and site now expose exact weighted progress so small M1 coverage gains do not disappear. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 11:14 PM EDT implementation pass continued M1 and tightened the progress calculation itself. The queue-only sync consumed violin 1 windows *1170-1260 and *1260-1350; the first became violin-positive source evidence and the second became checked non-playing/unclear evidence. The deployed active-practice scan folded both into the coverage ledger. Live state now reports 96 sample results, 85 active-violin samples, 11 checked non-violin samples, 283 active intervals, 2 h 16 min checked video, 2 h 7 min measured active practice, and a 199 h 19 min provisional archive

estimate. The tracker implementation now sums unrounded gate points before formatting the public label, so this small but real coverage gain moves the expected exact implementation label to 31.128 percent after deployment. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 11:38 PM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *1350-1440 and *1440-1530; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31.131 percent exact weighted implementation completion, 2 h 19 min checked video out of 213 h 23 min, 2 h 10 min measured active practice, 98 sample results, 87 active-violin samples, 11 checked non-violin samples, 294 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 11:55 PM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *1530-1620 and *1620-1710; the first became checked non-playing/unclear evidence and the second became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.134 percent exact weighted implementation completion, 2 h 22 min checked video out of 213 h 23 min, 2 h 12 min measured active practice, 100 sample results, 88 active-violin samples, 12 checked non-violin samples, 300 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 12:09 AM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *1710-1800 and *1800-1890; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.136 percent exact weighted implementation completion, 2 h 25 min checked video out of 213 h 23 min, 2 h 15 min measured active practice, 102 sample results, 90 active-violin samples, 12 checked non-violin samples, 313 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 12:35 AM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *1890-1980 and *1980-2070; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.139 percent exact weighted implementation completion, 2 h 28 min checked video out of 213 h 23 min, 2 h 18 min measured active practice, 104 sample results, 92 active-violin samples, 12 checked non-violin samples, 322 active intervals, and 250 queued windows. This is real practice-time coverage progress

only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 03:00 AM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *2070-2160 and *2160-2250; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.142 percent exact weighted implementation completion, 2 h 31 min checked video out of 213 h 23 min, 2 h 21 min measured active practice, 106 sample results, 94 active-violin samples, 12 checked non-violin samples, 331 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 05:11 AM EDT source-score acceleration pass advanced M2 and M4 instead of continuing only the slow archive scan. The implementation added a local MusicXML source excerpt for the Scherzo-Tarantelle solo-violin opening motif, wired it into the Wieniawski score target, and changed the symbolic matcher so this source target can accept a four-note measure-level run with at least three distinct pitch classes. The parser now preserves written flats such as B-flat for staff placement while still matching their pitch class to detected audio, which closes the failure mode where a normalized A-sharp could render on the wrong staff position. The score snippet renderer now receives the target key signature and renders a G-minor two-flat key signature before the matched notes. The completion tracker now separates `acceptedMeasureMatchCount` from `longPhraseAcceptedCount`, so Curtis can show one verified measure-level score/audio match without claiming solved long-phrase transcription. The regression suite increased to 103 tests and now covers source-backed Wieniawski symbolic notes, four-note source-measure matching, written-flat staff placement, accepted-measure counting, and the continued rejection of one-note visual crops. This pass does not solve long-phrase transcription; it creates the first narrow, test-covered lane from source score notes to a grouped score–audio–video–transcription match.

The remaining M1 work is continuing the low-cost media pass over every unmeasured strict-ledger video in chronological windows, then replacing the sample-biased estimate with full-archive checked intervals. The remaining M10 work is the larger benchmark suite for repeated notes, arpeggios, fast runs, score-free exercises, score-coordinate truth, accepted notation/media completeness, and practice-time accounting regressions.

The long-phrase track should now run in parallel with full-archive scanning. Waiting for all 234 h 57 min of archive coverage before attempting a single measure would be too slow. The faster strategy is a narrow accepted-measure lane:

1. select one active violin-positive window from an already captured practice day, starting with 2026-05-03 because it already has hidden detected note series; 1213
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2. obtain or manually verify the corresponding symbolic Scherzo-Tarantelle measure from a source-backed score; 1215
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3. run the existing pitch-class phrase matcher against the stored detected series; 1217
4. accept the group only if the exact score notes, displayed transcription, local audio, and local video all agree; 1218
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5. then extend the same source-backed score map outward to adjacent measures and longer phrases. 1220

The next coding pass should: 1221

1. verify one symbolic Scherzo-Tarantelle measure with exact pitch names and measure context; 1222
2. feed that measure into the existing symbolic-score target path; 1223
3. mine the stored 2026-05-03 detected note series for a contiguous score/audio match; 1224
4. accept at most one compact grouped score-clip-audio-transcription pair if the evidence gates pass; 1225
1226
5. keep running queue-only owner-media sync in parallel, starting with violin 1 at *2250-2340; 1227
6. rerun the active-practice scan after those uploads and record checked, active, and pending totals. 1228

Curtis can now pursue the first verified measure before M1 is complete, but it still cannot claim a score match from guessed scanned crops, one-note anchors, or reference-audio pitch traces. The speed-up is to shrink the musical target to one verified measure and make that one measure pass the same gates that a future long phrase will need. 1229
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6 Materials and methods 1233

Backend 1234

The backend is a FastAPI service deployed on Railway. It exposes <https://curtis.aolabs.io/api/curtis/media-status> for compact state, <https://curtis.aolabs.io/api/curtis/> 1235
1236

ops—check for fuller operational state, local stored sample media for embedded playback after the media is violin-positive, sample indexes for duplicate-window avoidance, and run endpoints for scans, media probing, analysis, and coaching. Persistent runtime state stores sources, inventory, review findings, media samples, analysis runs, and scan history.

Source scan

The default public source is <https://www.youtube.com/@nalalan>. The scanner queries public video metadata, classifies videos by title, duration, and candidate reasons, and stores inventory records. Dated practice logs and long-form videos are treated as practice candidates. The media probe now keeps one early baseline window but then probes deeper quarter, midpoint, five-eighths, three-quarter, and tail windows before spending additional passes on early setup probes, and each stored sample is keyed by video identifier plus window start so later passes can expand coverage without overwriting earlier evidence. When configured above the dense-sampling threshold, the probe and owner-media helper generate dense 90-second windows from the start of each video through the latest valid window so the full practice archive can be checked without the earlier retention cap. Each sample is then classified by the violin-presence sampler before it can count as active practice evidence. The owner-media helper also checks its local cache of already captured browser samples, reuses classifier results when the file signature and classifier version match, prioritizes active-scan pending windows before generic expansion or blind sampling, can run in queue-only mode to consume chronological pending windows without falling back to unrelated cached positives or blind windows, resumes cached queue windows after an interrupted capture, includes cached negative queue windows so non-playing time is checked rather than ignored, prioritizes unsynced violin-positive cached windows during normal mode before capturing more blind windows, and expands around already-found violin-positive windows before returning to blind sampling. The practice-hours total uses a stricter title-confirmed ledger from violin 1 onward: violin-numbered videos and date-titled logs count toward the top-line session duration, while unrelated long videos remain indexed but outside the practice-hours total. The scan does not itself inspect performance quality.

Model review

The model layer reviews sampled media sections where usable excerpts exist. GPT-5.5 is used for text and visual review, and separate audio models are configured for audio review and piece-title

verification. Findings are sanitized so weak evidence terms such as obscured, not audible, or no clear evidence produce an unjudged state rather than an unsupported critique. Piece labels are also sanitized: coach reviews can store candidate evidence but cannot create confirmed repertoire titles, and the piece-identification layer must provide source-window evidence plus independent verification before it updates the current piece list. Same-video consensus can compare multiple sampled windows before accepting a title; for long date-stamped practice captures with later samples, setup or non-playing windows are excluded from confirmation, and confirmed active titles are versioned so older labels are rechecked when source rules change. User-rejected false labels are excluded from later passes only for the source where the correction applies, so later videos can still identify that repertoire when present. A source correction also invalidates older confirmed labels for that same source instead of promoting the next stale candidate, and repeated corrected false labels keep that source out of the confirmed repertoire list until a stronger acceptance path exists. Rejected proposed titles, top candidates, and verification titles are scrubbed from current API output, not only from confirmed piece records. The 5/1 forensic pass rejected prior Paganini, Bruch, Mendelssohn, Brahms, Sibelius, Tchaikovsky, Wieniawski, Saint-Saens, Ravel, Bazzini, Ernst, Carmen, Zigeunerweisen, Bach Partitas, Kreisler Praeludium and Allegro, Sarasate family labels, Ysaye Sonata No. 3 Ballade, Mozart K. 216, Glinka Ruslan and Lyudmila, Strauss Till Eulenspiegel, Strauss Don Juan, Smetana Bartered Bride, Prokofiev Classical Symphony, Dvorak Carnival Overture, Shostakovich Symphony No. 5, Ravel Bolero, the first broad orchestral candidate batch, and the later overture, dance, and pops-orchestral candidate batch for that source. Alan then supplied the ground-truth source label: Haydn Symphony No. 94, IV. Finale, Violin I part. The system now treats that label as human-confirmed for 5/1, stops injecting new seeded guesses for the same source, and keeps Curtis-level completion unscored until judged playing evidence supports a score. Alan also supplied the 5/2 source label, Wieniawski Scherzo-Tarantelle, Op. 16, and then confirmed that all violin footage on 5/3 was the same piece. The training state now records all three source labels as confirmed sources and reports zero score-aligned windows until pitch/rhythm extraction has matched a source window to a score section. Transcribed samples can now produce learned pitch/rhythm fingerprints only after their media window is violin-positive, and visible notation requires note/rhythm verification. Future samples can receive a nearest-source hint when their extracted notes and rhythms resemble a prior confirmed source, an explicit title-labeled calibration upload, a symbolic reference, or a public labeled YouTube reference indexed through the official Data API; if the extracted pattern does not belong to a known score, it remains piece-or-exercise pending rather than being forced into repertoire. Public reference metadata can improve candidate

vocabulary and target selection, but it does not become an audio fingerprint until a permitted media source exists. Possible labels and model-only verified titles remain uncertain daily evidence unless a source label is confirmed, so future misses appear as missing evidence rather than as wrong repertoire. Piece-identification output does not create a Curtis-level readiness percentage by itself; readiness scoring requires judged playing evidence. Dated practice entries are keyed by the source-video date rather than the later processing time, so historical uploads cannot masquerade as today's practice state. The owner-browser sync uses the sample index to prioritize uncaptured public YouTube windows rather than re-uploading already reviewed excerpts, and the piece-identification pass can test multiple active moments inside each captured sample before withholding the title. The daily-record builder joins the strict `violin 1` ledger to violin-positive media samples, hidden machine note/rhythm events, practice-time evidence, clips, score packet support, repeated-fragment or repeated-pattern heat maps, score-aware read fields, and observation records. Repertoire entries are built only from confirmed daily-record evidence and keep progress percentages unassigned until clips, notation, and score or pattern alignment support them.

Progress ledger

AO Progress tracks Curtis as part of a larger multi-application state record. The first progress snapshot included Curtis home, Curtis media state, other AO Labs apps, Imagineer state, Imagineer paper, Relay state, and public project surfaces. Curtis contributes dated practice inventory, current focus, review status, and future trend signals.

7 Limitations

The current system cannot predict acceptance at Curtis. It also cannot replace a musician, teacher, jury, studio lesson, or audition process. It can be wrong when the source media are incomplete, when audio is unavailable, when video frames hide the physical action, or when model review fails. The broad micro-transcription gate is currently held closed: candidate note runs can be stored, but no displayed passage transcription should be treated as solved until its notes audibly match the paired clip. The only current displayed notation is two strict audio-checked note fragments, so the useful claim is limited: Curtis Media Review turns practice media into a persistent, source-bounded record that can support iteration over time.

References

1.	K. A. Ericsson, R. T. Krampe, C. Tesch-Romer, The role of deliberate practice in the acquisition of expert performance. <i>Psychological Review</i> 100 , 363–406 (1993).	1326
2.	A. Williamon, Ed., <i>Musical Excellence: Strategies and Techniques to Enhance Performance</i> (Oxford University Press, Oxford, 2004).	1327
3.	A. N. Pham, <i>Curtis Media Review</i> , Live software system, curtis.aolabs.io , 2026.	1328
4.	A. N. Pham, <i>AO Progress Ledger</i> , Live longitudinal state system, progress.aolabs.io , 2026.	1329